

AYUSH SINGH

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Education

M.Tech in Artificial Intelligence

Indian Institute of Science, Bangalore

2022 – 2024

CGPA: 7.6/10

B.Tech in Electronics and Electrical Engineering

KIIT University, Bhubaneswar

2015 – 2019

CGPA: 8.81/10

Experience

Applied AI Researcher - II, Fujitsu Research, India

Jul 2024 – Present

Retrieval-Augmented Video Understanding with Compositional Reasoning over Graphs

WACV 2026 ↗

- Developed RAVU, a retrieval-augmented video understanding framework that models long videos as spatio-temporal graph memory, enabling LMMs to answer temporal and causal VideoQA queries using only 5–10 retrieved frames.
- Engineered compositional graph reasoning over entity tracks and event-level memory, improving temporal VideoQA by up to 12.1% over CLIP/text-based retrieval baselines; research PoC was later leveraged in Fujitsu's Arena Agent for worker-safety video analysis.

Building Compliance Paralegals via Clause Graph Reasoning over Contracts

EACL 2026 ↗

- Developed COMPACT, an EACL 2026 accepted legal reasoning framework that models contracts as clause graphs to capture cross-clause dependencies across obligations, exceptions, definitions, and temporal conditions.
- Built ACE, a 4.7K-scenario compliance benchmark from 633 real-world contracts, and fine-tuned compact LLMs via reasoning distillation to improve multi-clause compliance accuracy by up to 43% points.

Research Project - Powered by Shell

Enhancing Context Length in Large Language Models

(Aug'23 – Jul'24)

Guided by Prof. Prasanta Kumar Ghosh, IISc and Avanindra Singh, Shell, Bangalore

- Collaborated with Shell's lead LLM team to enhance the context length of large language models using parallelization and optimization techniques to strike a balance between computational efficiency and memory usage.
- Explored various techniques, including linear interpolation for rotary positional embeddings, and ongoing work includes implementing NTK-aware scaled rotary embeddings, Flash Attention, and ALiBi techniques to boost LLM capabilities.

Projects

Ad Click Prediction | Personal Project

(Jan'23)

- Leveraged Logistic Regression and XGBoost to predict click-through rate of ads for KDD Cup 2012, Track 2 dataset.
- Achieved an AUC score of 0.681 and MAPE of 6.19% on test data by hand-crafted feature extraction.

Recommender System | Personal Project

(Jan'23)

- Developed recommendation systems using Content-Based, Collaborative and Hybrid approaches for Movielens dataset.
- Improved RMSE by 4.3% over Matrix Factorization technique by incorporating Neural Collaborative Filtering.

Arithmetic Word Problem Solver | Prof. Shirish Shevade

(Aug'23)

- Used T5 Tokenizer and built a Transformer model from scratch for translating English text into arithmetic expressions.
- Achieved a 72% accuracy by evaluating generated postfix arithmetic expressions and obtaining a single numerical result.

Hateful Memes Identification with Multimodal Classification | Prof. Rajiv Soundararajan

(Apr'23)

- Developed and assessed three Multimodal architectures using ResNet, SBERT, R-CNN, BERT Tokenizer, VisualBERT, and CLIP on Facebook's Hateful Meme dataset.
- Achieved a peak AUC score of 0.67 by employing an early fusion technique to combine text and visual embeddings.

Text Classification using Graph Neural Networks | Prof. Shirish Shevade

(Sep'23)

- Implemented Text-GCN with a text graph based on word-word co-occurrence and document-word relations.
- Enhanced accuracy by 1.47% on the Movie Review dataset by integrating Graph Attention Networks within Text-GCN.

Image Generation Using Deep Generative Models | Prof. Prathosh A.P

(Sep'23)

- Utilized Variational Autoencoders with RMSE and KL Divergence loss to generate human faces on the CelebA dataset.
- Enhanced image quality with DCGAN compared to VAE and generated Bitmoji using the Cartoon dataset.

Multilingual Speaker Recognition | Prof. Prasanta Kumar Ghosh

(Apr'23)

- Utilized MFCC feature extraction to capture critical acoustic characteristics from self-collected raw speech signals, distinguishing ten bilingual speakers using KNN, CNN, and LSTM models.
- Explored the impact of noise and the suitability of VCV sentences, achieving a peak accuracy of 78%.

Relevant Coursework

- Speech Information Processing
- Computer Vision
- Pattern Recognition and Neural Networks
- Digital Video Perception
- Advanced Image Processing
- AI for Medical Image Analysis
- Data Structures and Algorithms
- Computational Linear Algebra
- Stochastic Models and Applications
- Computational Methods of Optimization

Technical Skills

Languages | Tools: Python, C, C++, Numpy, Pandas, ScikitLearn, PyTorch, TensorFlow, HuggingFace.

Domains: NLP, Speech/Image/Video Processing(including Computer Vision), Generative AI.

Accomplishments | Extracurriculars

- **Awarded the prestigious Shell MTech Fellowship** for academic excellence and research potential, 2022 - 2024.
- **Teaching Assistant** for course E0-251 Data Structures and Algorithms.
- Secured an All India Rank of 108 (99.88 percentile) in GATE 2020 for Electrical Engineering.
- Excellence Certificate in Data Structures and Algorithms in Python by Coding Ninjas.
- Educator at Top Test Series and Subject Matter Expert at Gate Academy (Unacademy) from 2020 to 2022.